Let $J = \int_{-\infty}^{\infty} dx \; e^{i(x^2-1)^2}$

The integrand has 3 saddle point: $x = \pm 1$ and $x = 0$. Suppose that J can be estimated by a local stationary phase approximation about the saddle points, i.e.

$$J = \sum_n^{\Gamma} J_n ,$$

where n corresponds to the saddle point. At each saddle point, write $x = z e^{i\theta} + (\text{saddle point})$. At $x = \pm 1$, we have

$$J_{\pm} = \int_{-\infty}^{\infty} dz \; e^{i\theta_{\pm}} \; e^{i\left[(z e^{i\theta_{\pm}} \pm 1)^2 - 1\right]^2}$$

and at $x = 0$ we have

$$J_0 = \int_{-\infty}^{\infty} dz \; e^{i\theta_0} \; e^{i(z^2 e^{2i\theta_0} - 1)^2}$$

To each saddle point approximation corresponds a weight:

$$W_0 = e^{i\theta_0} \; e^{i(z^2 e^{2i\theta_0} - 1)^2} , \quad W_{\pm} = e^{i\theta_{\pm}} \; e^{i\left[(z e^{i\theta_{\pm}} \pm 1)^2 - 1\right]^2}$$

such that $J \approx \langle W_0 \rangle + \langle W_+ \rangle + \langle W_- \rangle$. I propose training three neural networks to learn three different angles for three seperate Monte Carlo integrals.

This might get you closer to the full result, but I anticipate 2 problems:

1) The true thimble is a curve, not three lines. I need to think more about how to learn the full contour...

2) The linear approximation I've proposed should probably be a weighted sum. If that's true, I suggest writing

$$J \simeq \alpha \langle W_0 \rangle + \beta \left(\langle W_+ \rangle + \langle W_- \rangle \right)$$

and finding α and β by fitting to Ross' solution. I'm not sure what α and β should be a priori, but I expect $\alpha \ll \beta$. Incidentally (if I understand your method correctly), $\langle W_0 \rangle$ is what you've calculated so far).